Tree trunks & bark

Continuous trunks, improved wrap shading, and five rebuilt bark materials.



Reported in the old shortcut build



Actual forest in the updated renderer · cropped

Different scene and camera. The close material comparisons on the next page use matched views.

Trunk seams: includes the continuous-ring trunk implementation previously staged in the candidate. Smooth taper normals and exact cylindrical tangents address lighting splits at UV wraps.

Bark: source colours retain detail through linear-light exposure processing. Tile boundaries and normal-map mip levels are corrected during authoring. One fissured material uses a generated source; four retain existing photographed scans.

Delivery: the shortcut selects the verified candidate at handoff. An already running old session needs to be relaunched to show the changes.

Matched material views

Production tree geometry and shader, with the same specimen and camera before and after the material pass.

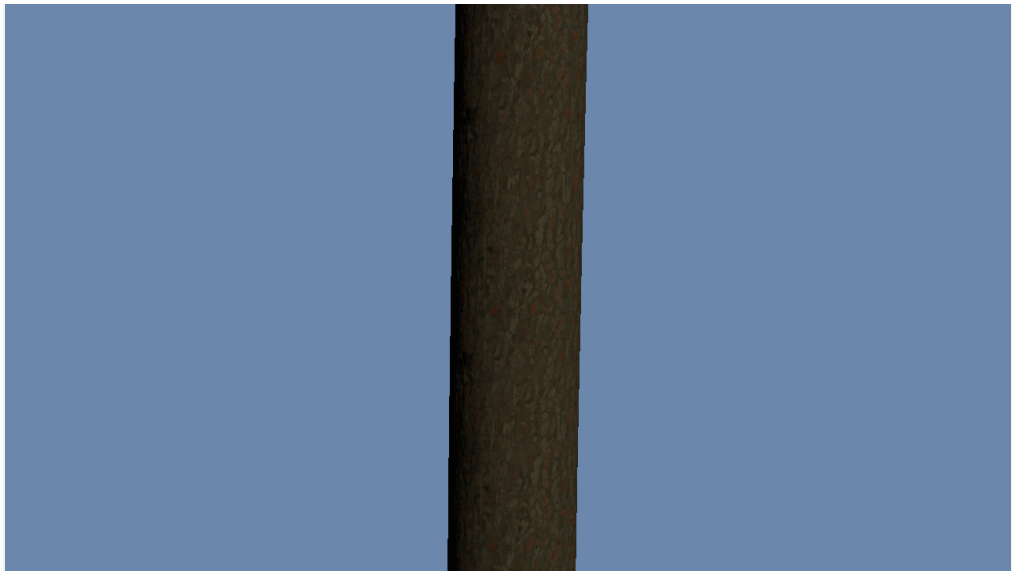
BEFORE MATERIAL PASS



UPDATED



Fissured bark: generated albedo with approximate derived relief, replacing the earlier washed-out material.





Scaly bark: photographed surface with retained colour and controlled repeat boundaries.

Validation & rendering budget

The changes run in authoring/model generation; the foliage shader and its texture-fetch count are unchanged.



Smooth · scan



Fissured · generated



Scaly · scan



Fibrous · scan



Peeling · scan

11 forms × 2 LODs

Wrap normal/tangent checks pass

Same triangle counts

Compared with the staged tree build

3.33 MiB bark bank

Same five512px BC7 material pairs

Native checks pass

Packaged tree and forest shaders

Published bark pixels passed **330,241** boundary, unit-normal and linear-mip checks. Native scratch-buffer checks ensure generated meshes own their data after cached buffers are reused.

Performance scope: no new draw passes, texture layers or per-frame processing. A live FPS comparison was not recorded while the existing WorldGen session was running; unchanged budgets are not a global FPS guarantee.

Limits: the generated layer's relief is inferred from colour, not measured geometry. Branch junction shapes remain the existing procedural architecture. The report covers trunk wrapping and bark materials.

Unity — normals at UV seams

Adobe — light cancellation, tiling and normal inputs

Moisan2011 — periodic-plus-smooth image decomposition

Poly Haven — scan reuse terms and attribution

Generated master and full prompt: assets-source/vegetation/tree-surfaces/ai/fissured-bark-20261107/. Source assets are versioned in the verified GitHub asset release.